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Project Title:	Impact of Age, Genetics Variants and High LET Track Structure on Mammary Cancer Risk Estimates		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Task Description:	A key NASA objective is to obtain quantitative data defining the risk of carcinogenesis following galactic cosmic ray (GCR) exposure. Current radiation quality factors used to determine cancer risk for space radiation do not provide the accuracy needed, and debate exists as to the relevance of the experimental studies they're based upon. To formulate better risk estimates, key data is needed including how age, genetic susceptibility, and radiation quality affect carcinogenic potential in irradiation sensitive tissues. Towards this aim, we will use a novel resource of a large collection of normal mammary epithelial cell strains, from women aged 16 to 80, including many in the age of astronauts, to address these important questions. We will obtain quantitative data regarding radiation quality effects; ROS and DNA damage signaling, radiation-induced centrosome defects, and enrichment of stem cell populations using this cohort of strains from various aged individuals. In addition, we will also investigate genetic susceptibility, including how loss of p16 expression, a sporadic event in individual cells of normal women in vivo and often observed in breast		

tumors, affects radiation quality effects and centrosome aberration frequency. Additionally we will analyze strains containing a BRCA1 or an ATM heterozygote genetic background, representing ~0.32 and ~1% of the normal population respectively, to determine how these genetic variants may make them more sensitive to high LET exposures. These studies will provide critical data, which can be used to model how genetic susceptibility, age, and radiation quality interplay in defining the cancer risk of mammary cells to high LET radiation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The intent of this proposal is to determine how age, genetic susceptibility, and radiation quality modify the risk of high LET radiation-induced cancer. A cohort of mammary epithelial cell strains isolated from women of various age groups is being used for these studies. Centrosome studies will be performed using both a novel flow cytometry based assay, which provides a less subjective and more rapid way to define centrosome defects, as well as a standard microscopy-based method. Previous studies have revealed that changes in stem cell numbers are associated with an increased risk of cancer. Thus our studies will also assess how radiation exposure and radiation quality affect stem cell populations, and whether age plays a role in the differences observed.
Materials & Methods: Frozen vials of primary cells from each individual under study (18) were obtained from our collaborator. Very early passage (p=2) cells were initiated and expanded for subsequent experiments. For our studies we have chosen to use passage 4-5 to obtain the necessary numbers for experiments. Approximately forty p4 and forty p5 vials were frozen down for each individual to have enough stocks to perform all experiments in the upcoming years. In the past year's experiments at BNL (NSRL14A and NSRL14C) we have exposed cells to cesium, Fe 300, Si 93, and Ti 300 MeV/u to obtain three sets of exposed populations of three doses (0, 0.05, and 0.5 Gy). Each strain was cultured in T75 flasks at BNL, exposed to radiation, and then one to two days post exposure, cells were trypsinized off the flasks and frozen down and sent back to LBNL. At the home institution, cells were thawed and expanded from 9-14 days for determination of the levels of centrosome defects and stem cell population changes.

Brief summary of past year's research accomplishments: In the first year of the project 16 cell strains were cultured and frozen down for subsequent studies, and some preliminary exposures were performed including Si 93 MeV/u and Ti300 MV/u exposures. Unfortunately due to shipping issues some of the samples did not arrive in a timely manner as scheduled and we incurred some loss of samples. Thus we had to again regrow and expand a number of strains to have sufficient numbers to expose to all the intended radiation qualities. In addition, we added 2 more strains to the panel to improve statistics regarding an age effect. In this second year we had two NSRL runs, NSRL14A and NSRL14C, and exposed cells to Si93, Fe 300, and Ti 300 MeV/u and performed two Tandem experiments exposing samples to Boron and Fe ions. Cesium experiments were also performed in this second year. We are hoping to focus the final year on thawing all the exposed samples and growing them out to then score for centrosome defects and stem cell population changes.

Having just returned from NSRL14C run, we only have preliminary results from the prior run this year that will be reported on. Given most past studies have concentrated on early times post exposure we are attempting to define centrosome aberrations at a later time, weeks post exposure, to provide additional data regarding how radiation exposure persistently affects cells, which would be more useful in modeling cancer risk. Thus preliminary studies were needed to define the optimal day post exposure to assay for maximum centrosome effects with different strains from different aged individuals. The yield of centrosome aberrations on different days following exposure was studied and it was revealed that day 9 to 12 post radiation resulted in the greatest elevations in centrosome defects. Thus we have gone on to grow out the various strains we are comparing to both day 9 and 12 to determine how levels are affected by the age of the individual and the radiation quality of the exposure. In these studies we have also investigated how persistent phospho-signaling for DNA damage response proteins correlates with centrosome aberrations, and observed that certain phospho-proteins correlate with centrosome levels whereas others do not. The link between cell growth and high centrosome aberration levels was also studied. These studies revealed following radiation cells with centrosome aberrations cycle more slowly.

The link between radiation quality and centrosome aberrations was studied in one strain, but further studies are planned to define how this changes with other aged individual strains. Both FACS-based studies and microscopy scoring reveals evidence of a radiation quality dependent effect, but the degree of this effect appears dose dependent. We have also investigated whether cells from different phases of the cell cycle show differing amounts of centrosome aberrations, and noted that both G1 and G2 cells show similar numbers of centrosome defects, and thus for future work will center on G1 cells for our analysis.

The final item we were interested in understanding regarding centrosome aberrations is whether radiation exposure causes an age dependent effect in terms of the level of centrosome aberrations induced. Thus far our work does not allow us to completely determine this but it is suggestive that there may be some sort of increase in levels with age.

Given that a change in ploidy is also a marker of cancer, we have plans to investigate whether an increase in ploidy is observed with radiation and whether increased age of the individual the strain was obtained from affects the degree of ploidy observed. Preliminary results with a strain from one young individual (21 yo) reveals a subtle effect that increases with dose, however no radiation quality effect is noted (both cesium and Fe 300 MeV/u show a similar pattern). In the upcoming year we will investigate more strains and more radiation qualities, to define these changes with more certainty.

Stem cell studies: We have optimized staining for AldefluorTM, a flow-based marker that is known to specifically mark and identify cells having progenitor/stem cell potential. Based on a time course study, we have determined that staining with Aldefluor for 30 mins, and sample acquisition at 4°C within half an hour post staining, produces reproducible results. Stained strains were gated based on forward and side scatter, and live singlet populations were used for final analysis. For positive identification of ALDH bright cells, cells were treated with ALDH alone or with ALDH + DEAB (inhibitor of ALDH activity that serves as an inhibitor to set gates to positively identify the ALDH bright population) and processed immediately. Live-dead discrimination was carried out by counter staining cells with DAPI. Results were analyzed using BD FACS Vantage DIVA (Becton Dickinson.)

A subset of cells from various age groups were expanded to 14 days post radiation exposure, trypsinized, stained with ALDH alone, or along with its inhibitor DEAB. Our cesium studies reveal that the proportion of ALDH positive cells varies between individuals of different ages, and suggests an age-dependent trend. Two strains from the young

	individuals show lower initial levels and lower levels following dose than does the oldest strain studied (from a 46 yo, in the age range of typical astronauts). However, a strain from the youngest individual studied thus far (16 yo) showed intermediate levels. When viewing all the strains we note a dose-dependent increase in ALDH positive cells post cesium exposure with the highest number of ALDH positive cells observed post 0.5Gy exposure. A comparison of a smaller subset of strains exposed to a high LET ion (Ti300 MeV/u), revealed a higher increase in ALDH positive cells with dose as compared to cesium in two strains from younger individuals. The astronaut-aged strain did not show a dose-dependent increase in ALDH with high LET exposure, however this could be due to issues with cell culture and bears repeating. The oldest strain analyzed in these stem cell studies (65 yo), also shows a dose-dependent increase with high LET exposure, however an age dependent increase in stem cell populations is not uniformly observed. Additional experiments with more various aged strains are underway. We have also stained these strains with well-characterized cell surface markers to identify the impact of high LET radiation on the distribution of epithelial lineages and progenitors using flow cytometry. Some changes in luminal and myoepithelial populations were noted with radiation exposure and these studies will be continued in the subsequent year. This work when completed should aid in defining in a quantitative manner the impact of age and radiation quality on endpoints that have been linked to cancer, which should provide data for modeling the risk of mammary cancer from space radiation exposures.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Bailey SM, Costes SV, Doetsch PW, Dynan WS, Kronenberg A, Rithidech KN, Saha J, Snijders AM, Werner E, Wiese C, Cucinotta FA, Pluth JM. "Understanding cancer development processes following HZE particle exposure: Roles of ROS, DNA damage repair, and inflammation." Radiation Research. Accepted Oct. 13, 2014. , Oct-2014